A person needs to eat burgers. Each burger contains a count of calorie. After eating the burger, the person needs to run a distance to burn out his calories.

If he has eaten i burgers with c calories each, then he has to run at least $3^i * c$ kilometers to burn out the calories. For example, if he ate 3 burgers with the count of calorie in the order: [1, 3, 2], the kilometers he needs to run are $(3^0 * 1) + (3^1 * 3) + (3^2 * 2) = 1 + 9 + 18 = 28$.

But this is not the minimum, so need to try out other orders of consumption and choose the minimum value. Determine the minimum distance he needs to run. Note: He can eat burger in any order and use an efficient sorting algorithm.Apply greedy approach to solve the problem.

Input Format

First Line contains the number of burgers

Second line contains calories of each burger which is n space-separate integers

Output Format

Print: Minimum number of kilometers needed to run to burn out the calories

Sample Input

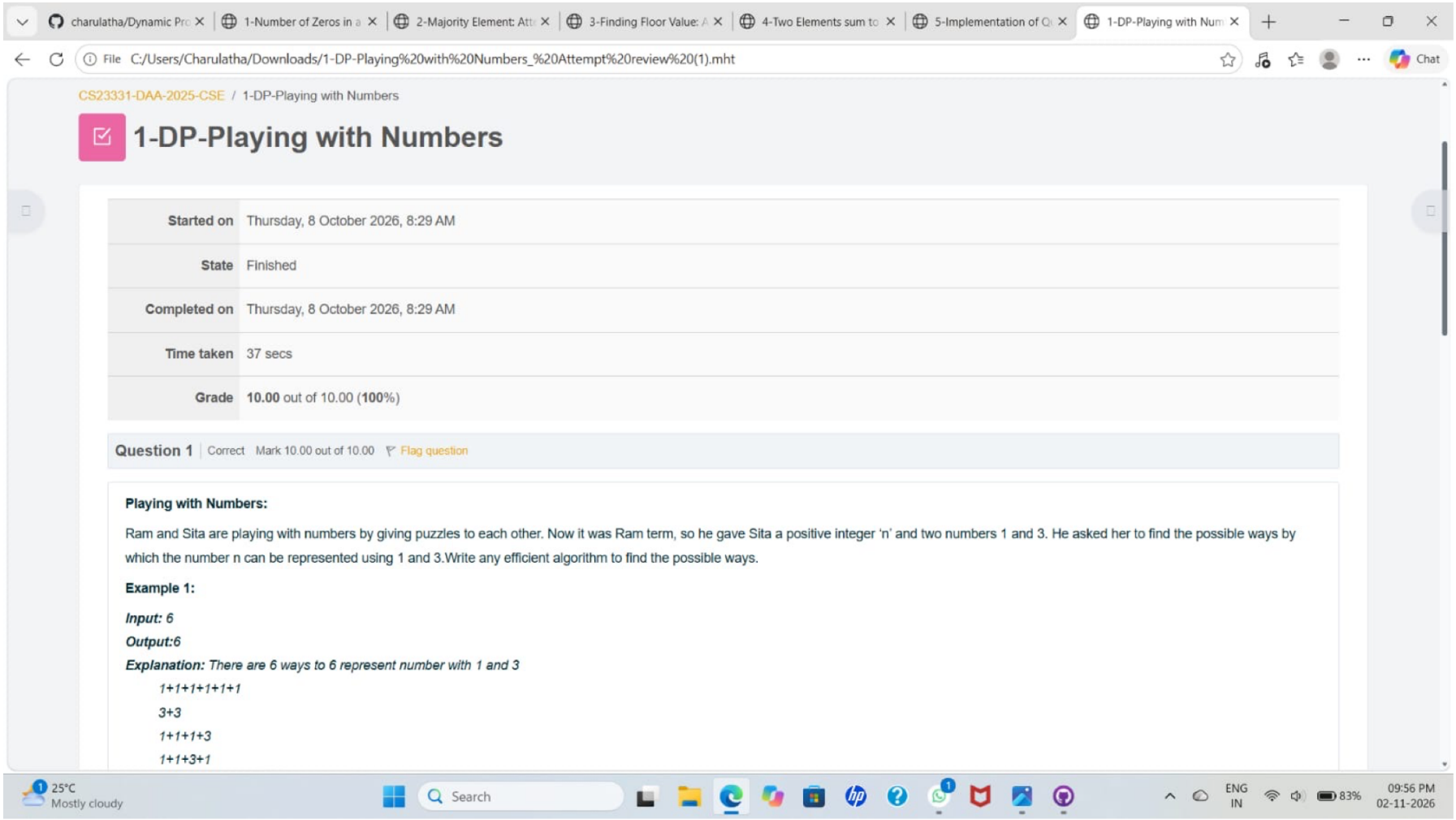
```
3
5 10 7
```

Sample Output

```
76
```

For example:

Test	Input	Result
Test Case 1	3 1 3 2	18


charulatha/Dynamic Pro
1-Number of Zeros in a
2-Majority Element: Att
3-Finding Floor Value: A
4-Two Elements sum to
5-Implementation of Qu
1-DP-Playing with Num
File C:/Users/Charulatha/Downloads/1-DP-Playing%20with%20Numbers_%20Attempt%20review%20(1).mht
Chat
CS23331-DAA-2025-CSE / 1-DP-Playing with Numbers
1-DP-Playing with Numbers
Started on Thursday, 8 October 2026, 8:29 AM
State Finished
Completed on Thursday, 8 October 2026, 8:29 AM
Time taken 37 secs
Grade 10.00 out of 10.00 (100%)
Question 1 Correct Mark 10.00 out of 10.00 Flag question
Playing with Numbers:
Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.
Example 1:
Input: 6
Output:6
Explanation: There are 6 ways to 6 represent number with 1 and 3
1+1+1+1+1+1
3+3
1+1+1+3
1+1+3+1
25°C
Mostly cloudy
Search
ENG
IN
83%
09:56 PM
02-11-2026

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*
Output:*6*
Explanation: *There are 6 ways to 6 represent number with 1 and 3*

1+1+1+1+1+1
3+3
1+1+1+3
1+1+3+1
1+3+1+1
3+1+1+1

Input Format

First Line contains the number n

Output Format

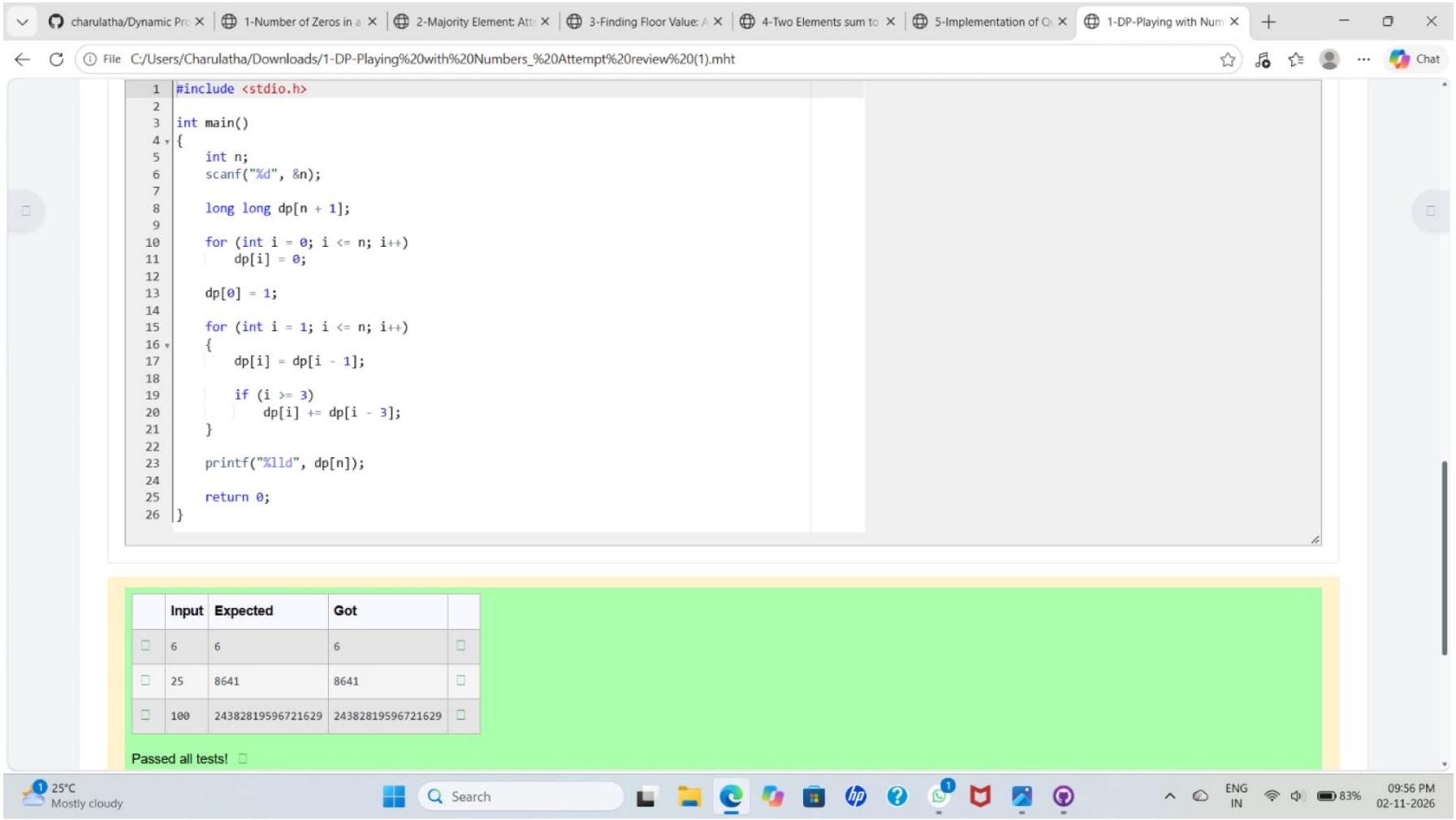
Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6


charulatha/Dynamic Pro
1-Number of Zeros in a
2-Majority Element: Att
3-Finding Floor Value: A
4-Two Elements sum to
5-Implementation of Q
1-DP-Playing with Num
File C:/Users/Charulatha/Downloads/1-DP-Playing%20with%20Numbers_%20Attempt%20review%20(1).mht
Chat
#include <stdio.h>
int main()
{
int n;
scanf("%d", &n);
long long dp[n + 1];
for (int i = 0; i <= n; i++)
dp[i] = 0;
dp[0] = 1;
for (int i = 1; i <= n; i++)
{
dp[i] = dp[i - 1];
if (i >= 3)
dp[i] += dp[i - 3];
}
printf("%lld", dp[n]);
return 0;
}
Input
Expected
Got
6
6
6
25
8641
8641
100
24382819596721629
24382819596721629
Passed all tests!
25°C
Mostly cloudy
Search
ENG
IN
83%
09:56 PM
02-11-2026

2-DP-Playing with chessboard

Started on	Thursday, 8 October 2026, 8:30 AM
State	Finished
Completed on	Thursday, 8 October 2026, 8:33 AM
Time taken	3 mins 8 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct | Mark 10.00 out of 10.00 | Flag question

Playing with Chessboard:

Ram is given with an n*n chessboard with each cell with a monetary value. Ram stands at the (0,0), that the position of the top left white rook. He is been given a task to reach the bottom right black rook position (n-1, n-1) constrained that he needs to reach the position by traveling the maximum monetary path under the condition that he can only travel one step right or one step down the board. Help ram to achieve it by providing an efficient DP algorithm.

Example:

Input

3

1 2 4

2 3 4

8 7 1

Output:

Question 1 | Correct | Mark 10.00 out of 10.00 | Flag question

Playing with Chessboard:

Ram is given with an n*n chessboard with each cell with a monetary value. Ram stands at the (0,0), that the position of the top left white rook. He is been given a task to reach the bottom right black rook position (n-1, n-1) constrained that he needs to reach the position by traveling the maximum monetary path under the condition that he can only travel one step right or one step down the board. Help ram to achieve it by providing an efficient DP algorithm.

Example:

Input

3

1 2 4

2 3 4

8 7 1

Output:

19

Explanation:

Totally there will be 6 paths among that the optimal is

Optimal path value:1+2+8+7+1=19

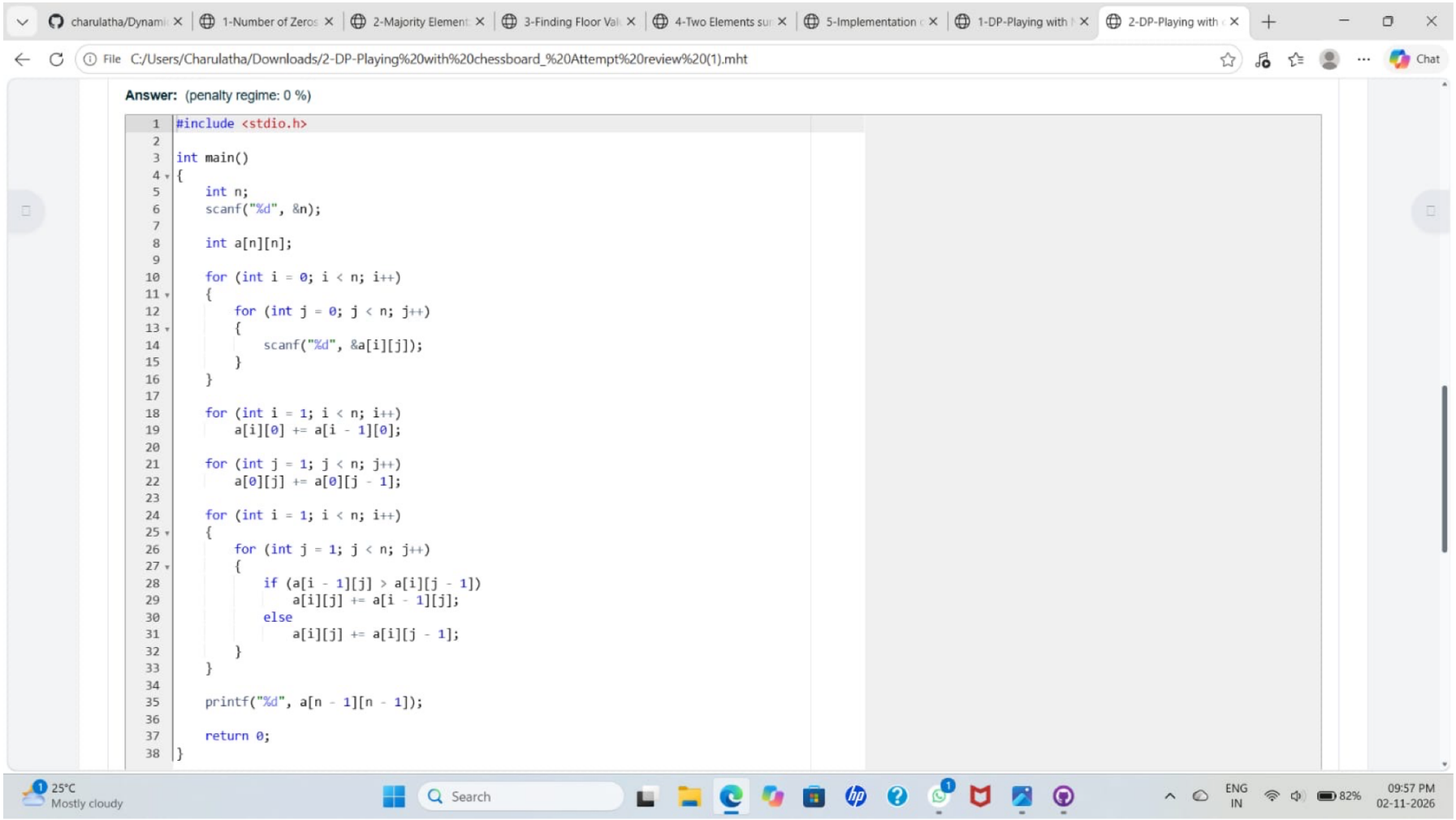
Input Format

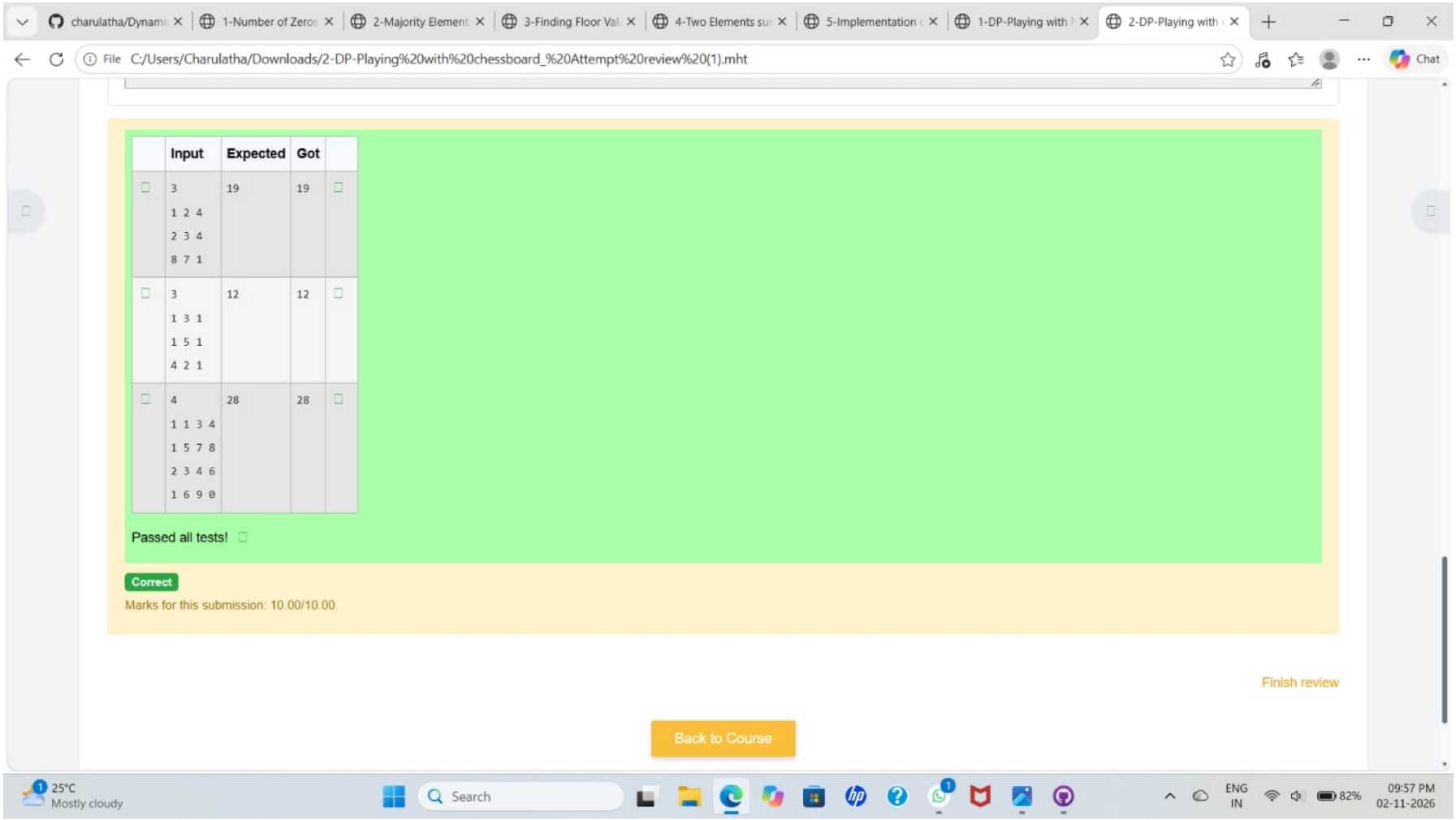
First Line contains the integer n

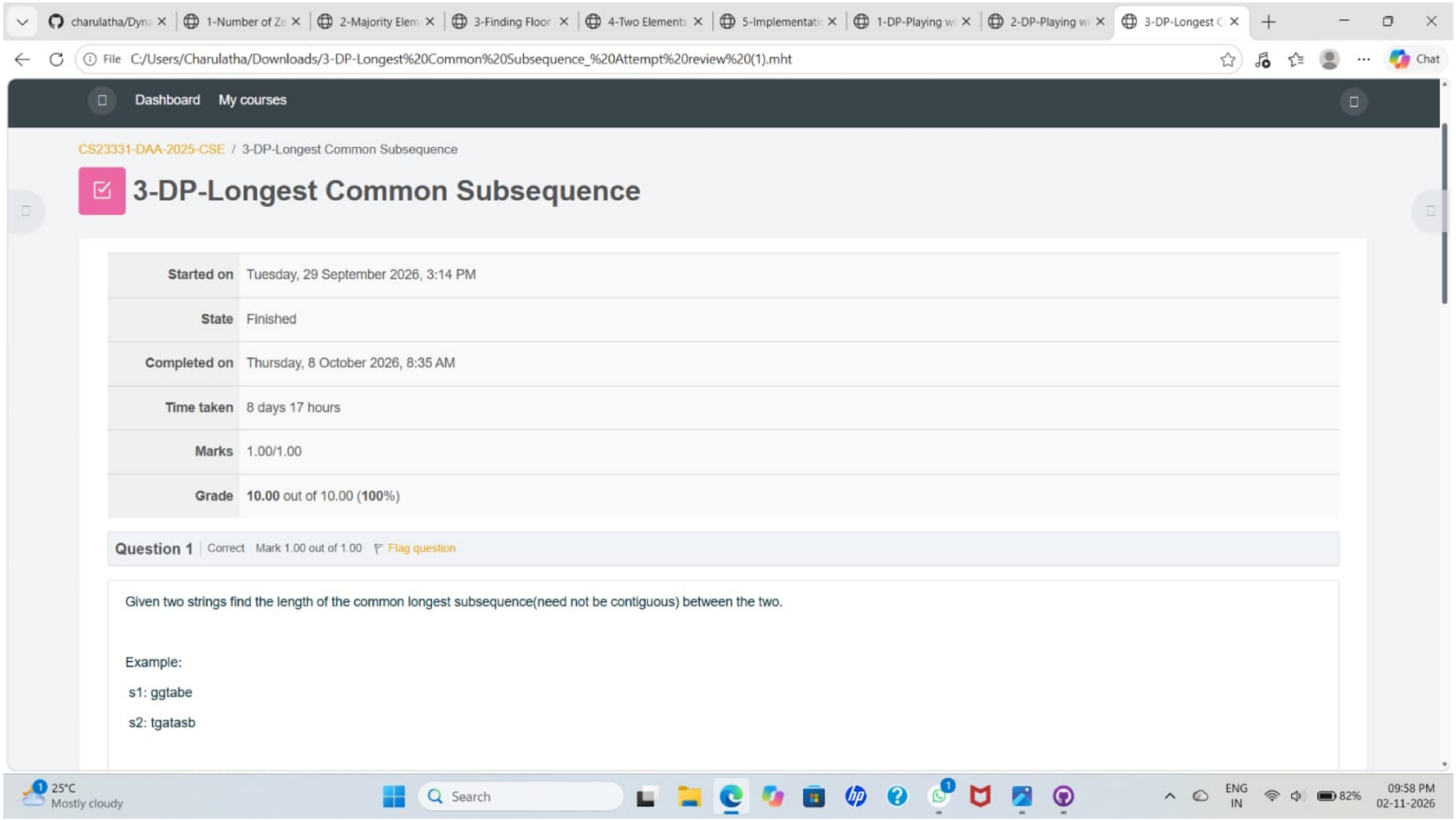
The next n lines contain the n*n chessboard values

Output Format

Print Maximum monetary value of the path


charulatha/Dynami
1-Number of Zeros
2-Majority Element
3-Finding Floor Val
4-Two Elements sur
5-Implementation
1-DP-Playing with
2-DP-Playing with
File C:/Users/Charulatha/Downloads/2-DP-Playing%20with%20chessboard_%20Attempt%20review%20(1).mht
Chat
Answer: (penalty regime: 0 %)
#include <stdio.h>
int main()
{
int n;
scanf("%d", &n);
int a[n][n];
for (int i = 0; i < n; i++)
{
for (int j = 0; j < n; j++)
{
scanf("%d", &a[i][j]);
}
}
for (int i = 1; i < n; i++)
a[i][0] += a[i - 1][0];
for (int j = 1; j < n; j++)
a[0][j] += a[0][j - 1];
for (int i = 1; i < n; i++)
{
for (int j = 1; j < n; j++)
{
if (a[i - 1][j] > a[i][j - 1])
a[i][j] += a[i - 1][j];
else
a[i][j] += a[i][j - 1];
}
}
printf("%d", a[n - 1][n - 1]);
return 0;
}
25°C
Mostly cloudy
Search
ENG
IN
82%
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02-11-2026


charulatha/Dynami
1-Number of Zeros
2-Majority Element
3-Finding Floor Val
4-Two Elements sur
5-Implementation
1-DP-Playing with
2-DP-Playing with
File C:/Users/Charulatha/Downloads/2-DP-Playing%20with%20chessboard_%20Attempt%20review%20(1).mht
Chat
Input
Expected
Got
3
1 2 4
2 3 4
8 7 1
19
19
3
1 3 1
1 5 1
4 2 1
12
12
4
1 1 3 4
1 5 7 8
2 3 4 6
1 6 9 0
28
28
Passed all tests!
Correct
Marks for this submission: 10.00/10.00.
Finish review
Back to Course
25°C
Mostly cloudy
Search
ENG
IN
82%
09:57 PM
02-11-2026


charulatha/Dyna
1-Number of Ze
2-Majority Elem
3-Finding Floor
4-Two Elements
5-Implementatic
1-DP-Playing wi
2-DP-Playing wi
3-DP-Longest C
File C:/Users/Charulatha/Downloads/3-DP-Longest%20Common%20Subsequence_%20Attempt%20review%20(1).mht
Chat
Dashboard
My courses
CS23331-DAA-2025-CSE / 3-DP-Longest Common Subsequence
3-DP-Longest Common Subsequence
Started on Tuesday, 29 September 2026, 3:14 PM
State Finished
Completed on Thursday, 8 October 2026, 8:35 AM
Time taken 8 days 17 hours
Marks 1.00/1.00
Grade 10.00 out of 10.00 (100%)
Question 1
Correct
Mark 1.00 out of 1.00
Flag question
Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.
Example:
s1: ggtabe
s2: tgatasb
25°C
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Search
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IN
82%
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02-11-2026

Question 1 | Correct | Mark 1.00 out of 1.00 | Flag question

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1	a	g	**g**	**t**	**a**	**b**	
s2	**g**	x	**t**	x	**a**	y	**b**

The length is 4

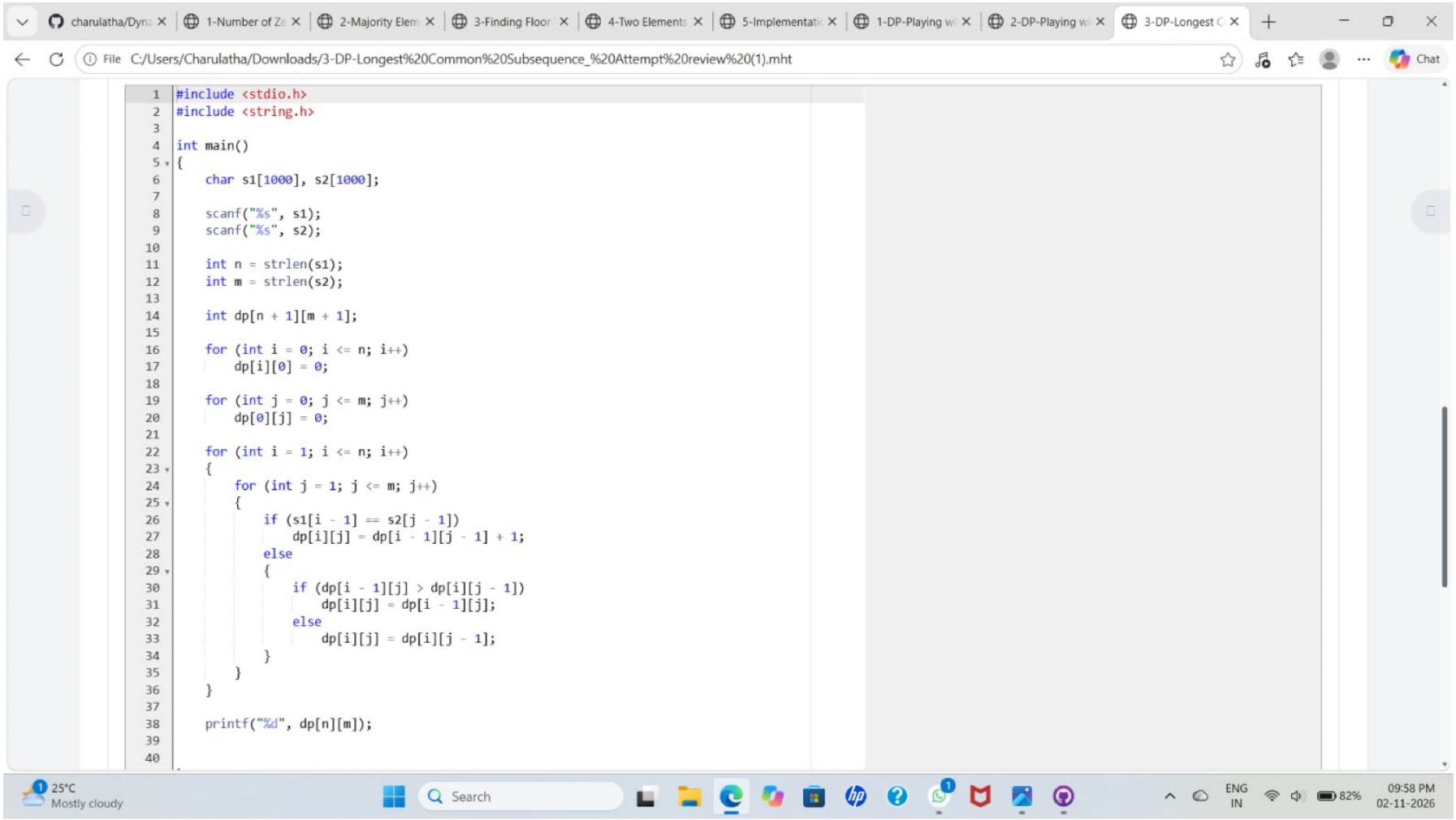
Solveing it using Dynamic Programming

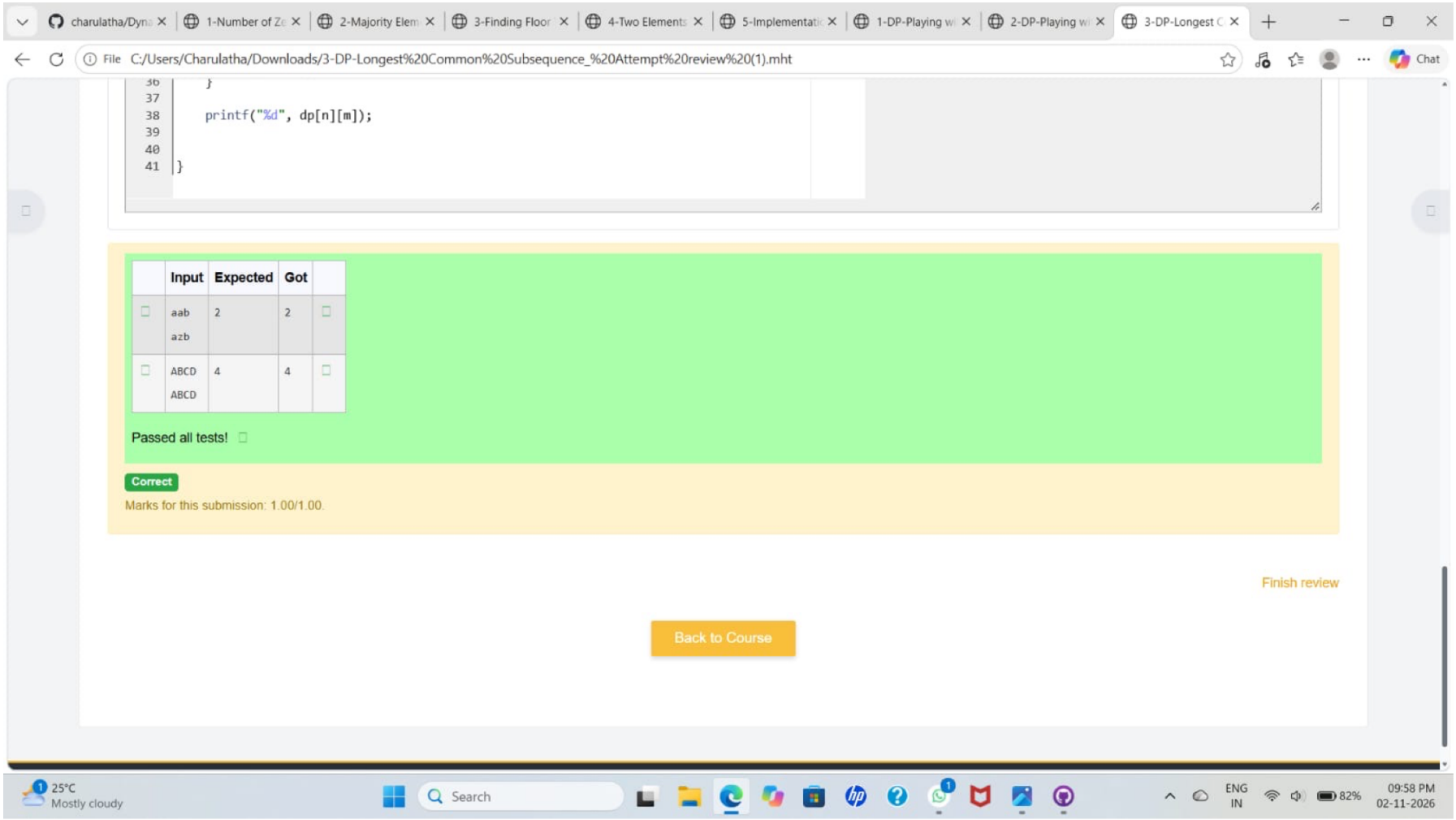
For example:

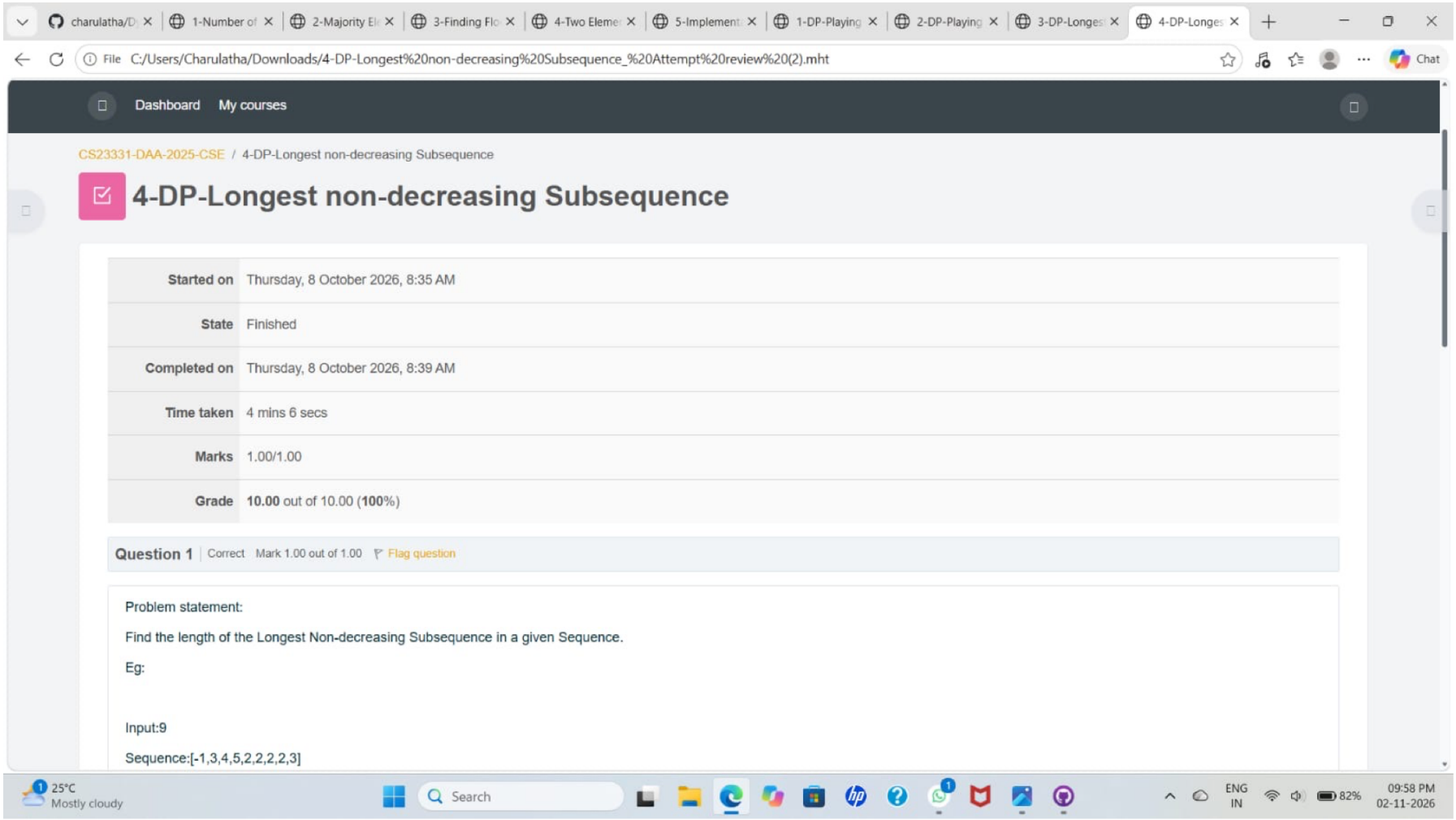
Input	Result
aab azb	2

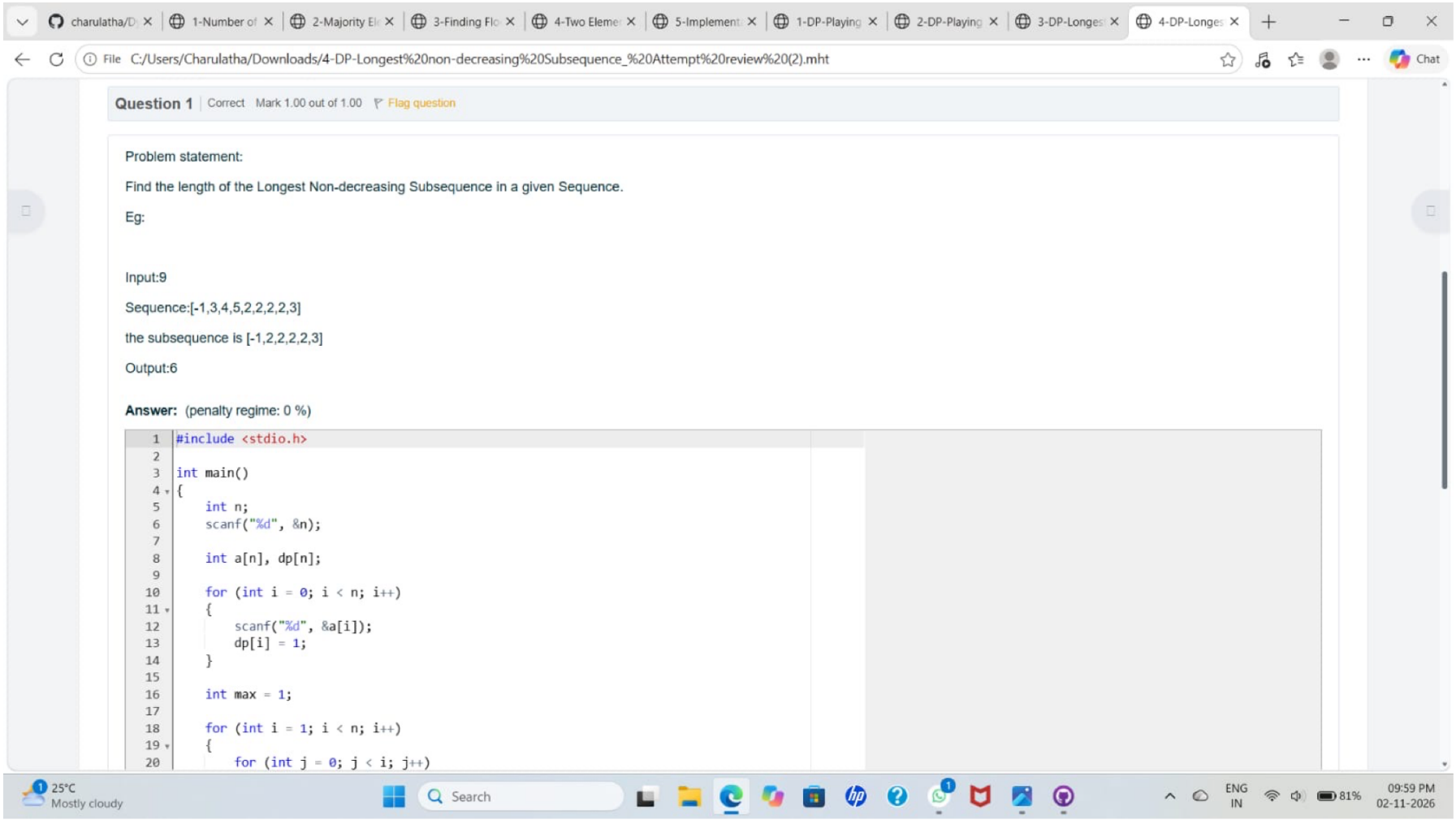
Answer: (penalty regime: 0 %)

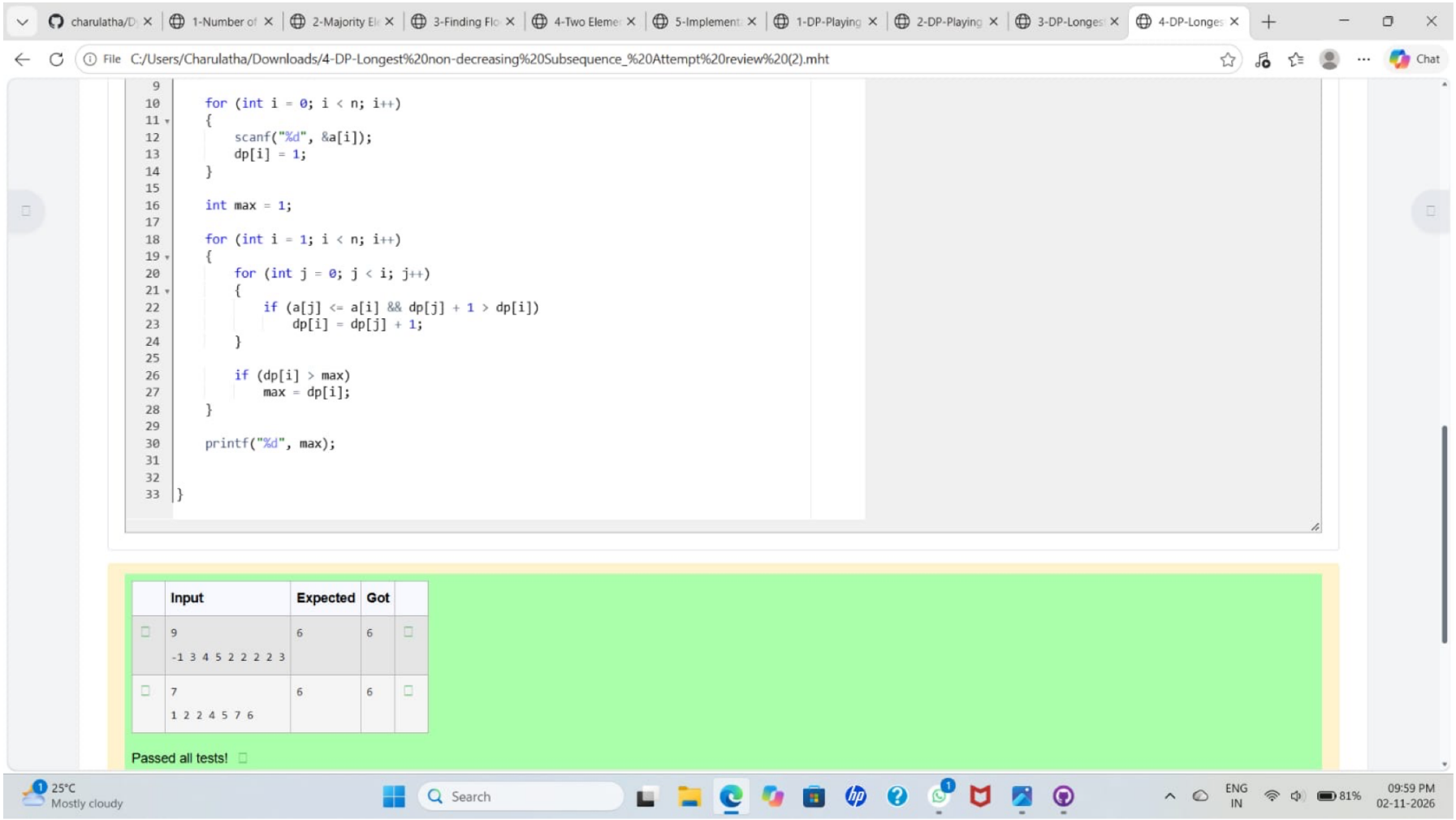
```
1 #include <stdio.h>
2 #include <string.h>
3
```


charulatha/Dyna
1-Number of Ze
2-Majority Elem
3-Finding Floor
4-Two Elements
5-Implementatic
1-DP-Playing wi
2-DP-Playing wi
3-DP-Longest C
File C:/Users/Charulatha/Downloads/3-DP-Longest%20Common%20Subsequence_%20Attempt%20review%20(1).mht
Chat
#include <stdio.h>
#include <string.h>
int main()
{
char s1[1000], s2[1000];
scanf("%s", s1);
scanf("%s", s2);
int n = strlen(s1);
int m = strlen(s2);
int dp[n + 1][m + 1];
for (int i = 0; i <= n; i++)
dp[i][0] = 0;
for (int j = 0; j <= m; j++)
dp[0][j] = 0;
for (int i = 1; i <= n; i++)
{
for (int j = 1; j <= m; j++)
{
if (s1[i - 1] == s2[j - 1])
dp[i][j] = dp[i - 1][j - 1] + 1;
else
{
if (dp[i - 1][j] > dp[i][j - 1])
dp[i][j] = dp[i - 1][j];
else
dp[i][j] = dp[i][j - 1];
}
}
}
printf("%d", dp[n][m]);
25°C
Mostly cloudy
Search
ENG
IN
82%
09:58 PM
02-11-2026


charulatha/Dyna
1-Number of Ze
2-Majority Elem
3-Finding Floor
4-Two Elements
5-Implementatic
1-DP-Playing wi
2-DP-Playing wi
3-DP-Longest C
File C:/Users/Charulatha/Downloads/3-DP-Longest%20Common%20Subsequence_%20Attempt%20review%20(1).mht
Chat
}
printf("%d", dp[n][m]);
}
Input
Expected
Got
aab
azb
2
2
ABCD
ABCD
4
4
Passed all tests!
Correct
Marks for this submission: 1.00/1.00.
Finish review
Back to Course
25°C
Mostly cloudy
Search
ENG
IN
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02-11-2026


charulatha/D
1-Number of
2-Majority El
3-Finding Flo
4-Two Eleme
5-Implement
1-DP-Playing
2-DP-Playing
3-DP-Longes
4-DP-Longes
File C:/Users/Charulatha/Downloads/4-DP-Longest%20non-decreasing%20Subsequence_%20Attempt%20review%20(2).mht
Chat
Dashboard
My courses
CS23331-DAA-2025-CSE / 4-DP-Longest non-decreasing Subsequence
4-DP-Longest non-decreasing Subsequence
Started on Thursday, 8 October 2026, 8:35 AM
State Finished
Completed on Thursday, 8 October 2026, 8:39 AM
Time taken 4 mins 6 secs
Marks 1.00/1.00
Grade 10.00 out of 10.00 (100%)
Question 1
Correct
Mark 1.00 out of 1.00
Flag question
Problem statement:
Find the length of the Longest Non-decreasing Subsequence in a given Sequence.
Eg:
Input:9
Sequence:[-1,3,4,5,2,2,2,2,3]
25°C
Mostly cloudy
Search
ENG
IN
82%
09:58 PM
02-11-2026


charulatha/D
1-Number of
2-Majority El
3-Finding Flo
4-Two Eleme
5-Implement
1-DP-Playing
2-DP-Playing
3-DP-Longes
4-DP-Longes
File C:/Users/Charulatha/Downloads/4-DP-Longest%20non-decreasing%20Subsequence_%20Attempt%20review%20(2).mht
Chat
Question 1 Correct Mark 1.00 out of 1.00 Flag question
Problem statement:
Find the length of the Longest Non-decreasing Subsequence in a given Sequence.
Eg:
Input:9
Sequence:[-1,3,4,5,2,2,2,2,3]
the subsequence is [-1,2,2,2,2,3]
Output:6
Answer: (penalty regime: 0 %)
#include <stdio.h>
int main()
{
int n;
scanf("%d", &n);
int a[n], dp[n];
for (int i = 0; i < n; i++)
{
scanf("%d", &a[i]);
dp[i] = 1;
}
int max = 1;
for (int i = 1; i < n; i++)
{
for (int j = 0; j < i; j++)
25°C
Mostly cloudy
Search
ENG
IN
81%
09:59 PM
02-11-2026


charulatha/D
1-Number of
2-Majority El
3-Finding Flo
4-Two Eleme
5-Implement
1-DP-Playing
2-DP-Playing
3-DP-Longes
4-DP-Longes
File C:/Users/Charulatha/Downloads/4-DP-Longest%20non-decreasing%20Subsequence_%20Attempt%20review%20(2).mht
Chat
for (int i = 0; i < n; i++)
{
scanf("%d", &a[i]);
dp[i] = 1;
}
int max = 1;
for (int i = 1; i < n; i++)
{
for (int j = 0; j < i; j++)
{
if (a[j] <= a[i] && dp[j] + 1 > dp[i])
dp[i] = dp[j] + 1;
}
if (dp[i] > max)
max = dp[i];
}
printf("%d", max);
}
Input
Expected
Got
9
-1 3 4 5 2 2 2 2 3
6
6
7
1 2 2 4 5 7 6
6
6
Passed all tests!
25°C
Mostly cloudy
Search
ENG
IN
81%
09:59 PM
02-11-2026